Skills Summary

Comparing Distributions and Compound Events

Use this reference while completing your worksheet or when studying.

Two questions decide every problem on the worksheet. For data: am I describing *where* the distribution sits (center) or *how spread out* it is? For chance: is this one event, or two stages happening together?

1. Comparing the centers of two distributions

Mean = (sum of the values) $\div$ (how many values). **Median** = the middle value of the *ordered* list.

With an **even** count there are two middle values: the median is halfway between them, so it need not be a value anyone actually got.

(!) A few unusually high or low values drag the *mean* toward them but barely move the *median*. When the two disagree, that gap is itself information about the distribution.

Example: Find the median of 4, 7, 9, 10, 12, 15.
Step 1. The list is already ordered and there are 6 values — an even count, so the median sits between the two middle ones rather than on a value.
Step 2. The 3rd and 4th values are 9 and 10, and halfway between them is $(9 + 10) \div 2$.
9.5

Try it: Find the median of 5, 8, 9, 13, 14, 19.

check: **11**

2. Comparing the spreads of two distributions

Range = greatest value $-$ least value. One pair of extremes decides it, so a single unusual value can double it.

Interquartile range = $Q_3 - Q_1$, the width of the middle half. Split the ordered list at the median; Q_1 is the median of the lower half and Q_3 is the median of the upper half. The extremes are deliberately left out.

Same center, different spread is the usual way two data sets differ. The smaller IQR belongs to the more consistent set.

Example: Find the IQR of 3, 5, 8, 10, 12, 14, 15, 19.
Step 1. IQR is about the middle half, so I split the ordered list in two and take the median of each half rather than using the extremes.
Step 2. Lower half $3, 5, 8, 10$ gives $Q_1 = (5 + 8) \div 2 = 6.5$; upper half $12, 14, 15, 19$ gives $Q_3 = (14 + 15) \div 2 = 14.5$; and $14.5 - 6.5$ is
8

Try it: Find the IQR of 2, 4, 6, 8, 12, 14, 16, 18.

check: **10**

3. Probability of a single event

$P(\text{event}) = \dfrac{\text{number of favourable outcomes}}{\text{number of equally likely outcomes}}$, always written in lowest terms.

Build the denominator before anything else: it counts *outcomes*, not categories. Four clubs with 40 students in them give a denominator of 40, not of 4.

Example: A jar holds 10 counters and 4 of them are yellow. One counter is drawn at random. Find $P(\text{yellow})$.

Step 1. One draw, so this is a simple probability: count the favourable outcomes over all the equally likely outcomes.

Step 2. 4 yellow out of 10 counters gives $\dfrac{4}{10}$, which reduces by a factor of 2.

$\mathbf{\dfrac{2}{5}}$

Try it: A jar holds 8 counters and 3 of them are green. Find $P(\text{green})$.

check: $\mathbf{\dfrac{3}{8}}$

4. Probability of a compound event

Two stages happening together: find each stage's probability and **multiply**. Adding gives the probability of *one or the other*, which is a different question.

Independent (spin then roll, or draw *with* replacement): the second probability is unchanged. **Dependent** (draw *without* replacement): recount the second stage from what is left — both the top and the bottom of the fraction shrink.

"At least one" is fastest through its opposite: $P(\text{at least one}) = 1 - P(\text{none})$.

Example: A fair coin is flipped and a number cube is rolled. Find $P(\text{heads and a } 6)$.

Step 1. Two stages happen together and the coin cannot change the cube, so the stages are independent and their probabilities multiply.

Step 2. $P(\text{heads}) = \dfrac{1}{2}$ and $P(6) = \dfrac{1}{6}$, so the product is $\dfrac{1}{2} \times \dfrac{1}{6}$.

$\mathbf{\dfrac{1}{12}}$

Try it: A spinner has 4 equal sections and 1 is blue. It is spun once and a number cube is rolled. Find $P(\text{blue and an even number})$.

check: $\mathbf{\dfrac{1}{8}}$